https://www.math.edu/~mgaerlan/

$(x^2+y^2-1)^2$

$4x(x^2+y^2-1)=0$

$4y(x^2+y^2-1)=0$

Critical points:

$\{(0,0)\} \cup \{(x,y)\in\mathbb{R}^2 \mid x^2+y^2=1\}$

$D = f_{xx}f_{yy} - (f_{xy})^2 = 48(x^2+y^2)^2 - 64(x^2+y^2)+16$

$f_{xx}+f_{yy}$

$48(1)^2 - 64(1)+16 = 0$

inconclusive

$$\frac{dx_1}{dt} = ax_1 + bx_2$$
$$\frac{dx_2}{dt} = cx_1 + dx_2$$

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{d\vec{x}}{dt} = A \vec{x}$$

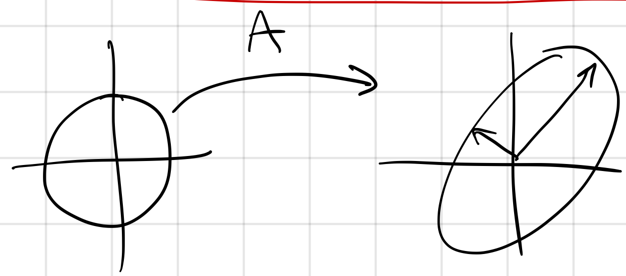
$$A\vec{v} = \lambda \vec{v}$$

$$\vec{x} = c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2$$

λ_1, λ_2 – eigenvalues

$\vec{v}_1, \vec{v}_2$ – eigenvectors

$$\boxed{\det(A - \lambda I) = 0}$$



$A - \lambda I$ ← identity matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}$$

$$A - \lambda I$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(a-\lambda)(d-\lambda) - bc = 0$$

$$ad - \lambda d - \lambda a + \lambda^2 - bc = 0$$

$$\lambda^2 - (a+d)\lambda + ad - bc = 0$$

$$\lambda^2 - (\operatorname{tr} A)\lambda + (\det A) = 0$$

$$\lambda^2 - \tau\lambda + \Delta = 0$$

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$$\tau = \operatorname{tr} A = a + d$$

$$\Delta = \det A = ad - bc$$

$$\lambda_1 + \lambda_2 = \tau$$

$$\lambda_1 \lambda_2 = \Delta$$

$(-)(-) = (+)$ $\quad (-)+(-) = (-)$

$(+)(+) = (+)$ $\quad (+)+(+) = (+)$

$(-)(+) = (-)$

$(+)(-) = (-)$

triangular

$$\begin{bmatrix} a & 0 \\ c & d \end{bmatrix} \text{ or } \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$$

lower triangular

upper triangular

$\lambda = a, d$

$$\frac{dx_1}{dt} = f_1(x_1, x_2)$$

$$\frac{dx_2}{dt} = f_2(x_1, x_2)$$

fixed point when

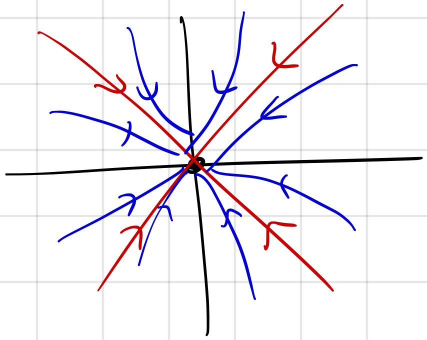
$f_1(x_1, x_2) = 0$

$f_2(x_1, x_2) = 0$

linear system fixed at $(0,0)$

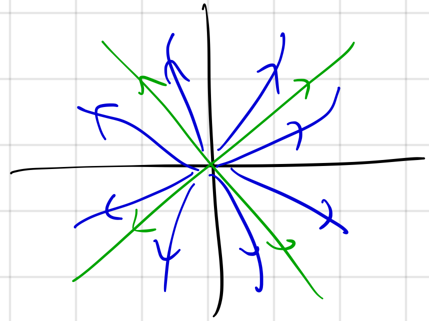
eigenvalues real

$\tau^2 - 4\Delta > 0$



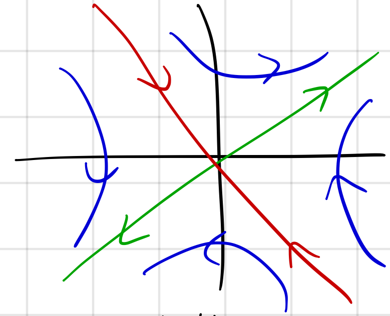
stable node

$\Delta > 0$
$\tau < 0$



unstable node

$\Delta > 0$
$\tau > 0$

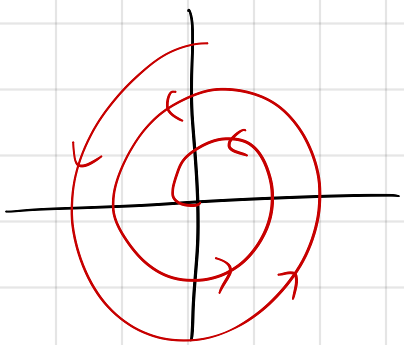


saddle node

$\Delta < 0$

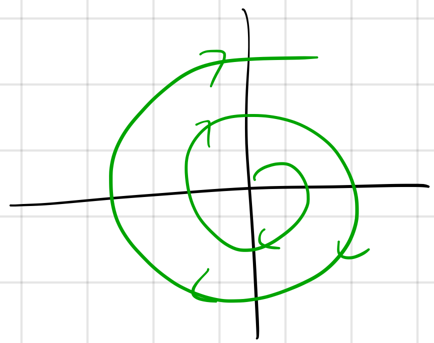
eigenvalues imaginary

$\tau^2 - 4\Delta < 0$



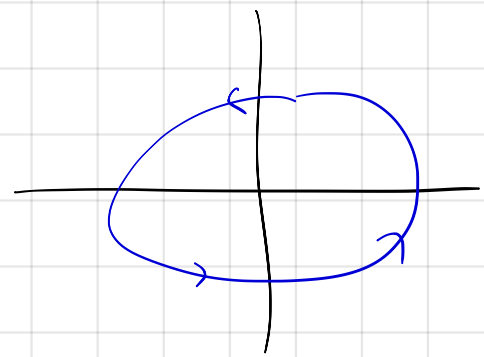
stable spiral

$\tau < 0$



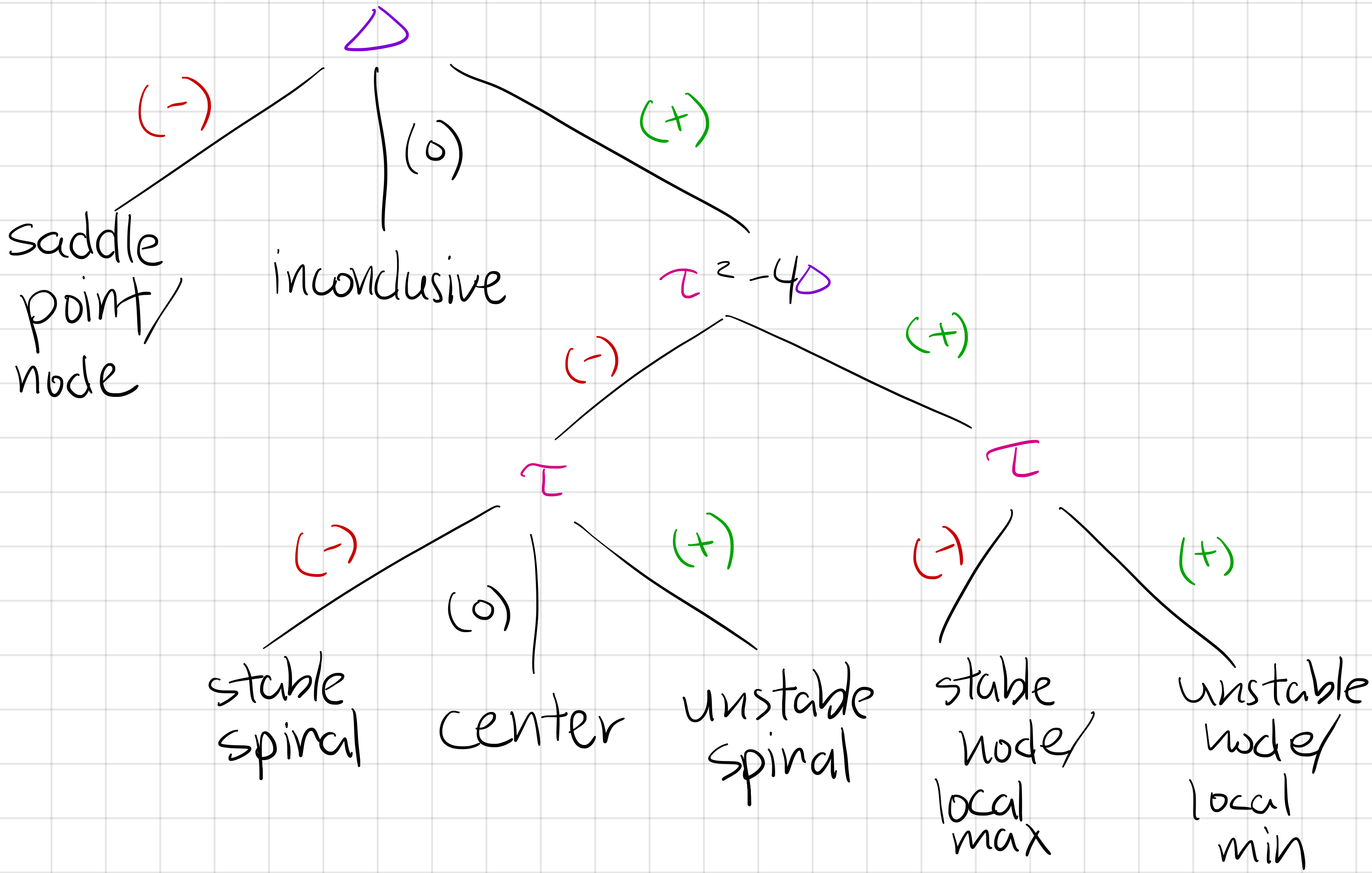
unstable spiral

$\tau > 0$



centers

$\tau = 0$


Δ
(-)
(0)
(+)
saddle point/node
inconclusive
τ²-4Δ
(-)
(+)
τ
τ
(-)
(0)
(+)
(-)
(+)
stable spiral
center
unstable spiral
stable node/local max
unstable node/local min

$$A = \begin{bmatrix} -3 & 3 \\ 6 & 4 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 3 \\ 6 & 4 \end{bmatrix} \quad \begin{aligned} \tau &= -3+4 = 1 \\ \Delta &= (-3)(4)-(3)(6) = -30 \end{aligned}$$

$$\lambda^2 - \lambda + 30 = 0$$

$$(\lambda - 6)(\lambda + 5)$$

$$\lambda = -5, 6$$

$$\vec{v}$$

$$\frac{dh}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$